

Homework #2 - MATH 261A

Noah Arias

Problem 1

We assume that $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$ and are I.I.D.

Part (a)

We know that $\bar{X}_n$ follows a Gaussian distribution, but we must determine the mean and variance parameters.

We do this by a straightforward expectation and variance calculation:

$$E(\bar{X}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n} \sum_{i=1}^n E(X_i)$$

Since each X_i follows a $N(\mu, \sigma^2)$, its mean is simply μ . Therefore, we can rewrite the last sum in the above calculation as:

$$\frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{\mu \cdot n}{n} = \mu$$

Therefore, we now know that the expectation of $\bar{X}_n$ is μ .

Now we calculate its variance in a similar manner:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

We note that the third equality is valid because since the X_i are *independent* as well as identically distributed. This means that any and all covariance terms are precisely zero. Continuing the calculation, we arrive at:

$$\frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2 \cdot n}{n^2} = \frac{\sigma^2}{n}$$

With this result, we can finally conclude that $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$.

Part (b)

The general structure of a confidence interval is following:

$$(\text{point estimate} \pm \text{critical value} \cdot \text{standard deviation/error})$$

We are specifically deriving a 95% confidence interval for the sampling distribution of $\bar{X}_n$ which we now know is $N(\mu, \frac{\sigma^2}{n})$.

The first thing we recognize is that μ is the parameter, whose location we are trying to ascertain with the estimate $\bar{X}_n$, for a given sample of size n . Thus, the point estimate for μ is $\bar{X}_n$.

Secondly, we must determine the critical value. This is dependent on the sampling distribution which is $N(\mu, \frac{\sigma^2}{n})$. Since we know not only our distribution is Normal but the population variance too, our a critical value is a z-score with form:

$$z_{\text{crit}} = \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}}$$

This expression follows a standard Normal distribution $N(0, 1)$.

The standard error is easy to find as it is simply $\frac{\sigma}{\sqrt{n}}$.

Starting at the requirement that for a 95% confidence interval, the following must be true:

$$P(-1.96 < \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96) = 0.95$$

To arrive at our desired confidence interval, we make take the inequality inside the probability operation and solve for μ :

$$\begin{aligned}
-1.96 &< \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96 \\
-1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \bar{x}_n - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
-\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< -\mu < -\bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}
\end{aligned}$$

Therefore:

$$P\left(\bar{X}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Part (c)

We now assume that there is another I.I.D. sample $X_{n+1}, \dots, X_{2n}$ of size n , that is also independent of the first sample, and is from another Normal distribution $N(\mu, \tau^2)$. (We also assume that τ^2 does **NOT** refer to the precision, but instead the variance of the second sample.)

Immediately, we can conclude that the sampling distribution of $\bar{X}_{2n}$ is yet another Normal distribution. Now, we must find the expectation and variance parameters, as before.

The expectation is straightforward after the following realization: when we calculated the expectation of the sample average, the sample size appeared in the numerator and the denominator of the fraction, which meant that it was cancelled out. This means that the expectation of this new statistic $\bar{X}_{2n}$ is also μ . So will any other statistic that involved taking the average of k number of variables.

The variance is best found by calculating it directly:

$$\text{Var}(\bar{X}_{2n}) = \text{Var}\left(\frac{\sum_{i=1}^{2n} X_i}{2n}\right) = \frac{1}{4n^2} \sum_{i=1}^{2n} \text{Var}(X_i) = \frac{1}{4n^2} \sum_{i=1}^{2n} \sigma^2 = \frac{\sigma^2 \cdot 2n}{4n^2} = \frac{\sigma^2}{2n}$$

Notice that the expression found for $\text{Var}(\bar{X}_{2n})$ is the exact same as $\text{Var}(\bar{X}_n)$ except that we replace n by $2n$. This is intuitive since all we did was double the sample size.

Therefore, to conclude $\bar{X}_{2n} \sim N\left(\mu, \frac{\sigma^2}{2n}\right)$